

A CONDITIONAL REDUCTION FOR THE HYPER-IDEAL QUESTION

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STATUS

This packet proves a conditional reduction. It does not prove the external Schur tensor-product dependency. Conditional status:

`conditional_reduction_likely_valid.`

SOURCE QUESTION

Botelho–Torres, *Hyper-ideals of multilinear operators*, arXiv:1504.00520, define the hyper-ideal $\mathcal{L}_{msc}$ of multilinearly sequentially continuous multilinear operators. They prove

$$\mathcal{CC} \circ \mathcal{L} \subseteq \mathcal{L}_{msc},$$

where $\mathcal{CC}$ is the closed ideal of completely continuous linear operators and $\mathcal{CC} \circ \mathcal{L}$ is the composition hyper-ideal.

Their Open Question 2 asks whether this inclusion is proper:

$$\mathcal{CC} \circ \mathcal{L} \subsetneq \mathcal{L}_{msc}.$$

The same discussion relates this question to the long-standing problem of whether the Schur property is preserved by projective tensor products.

CONDITIONAL DEPENDENCIES

The only external dependency is the following unproved assumption.

Schur tensor counterexample hypothesis. There exist Banach spaces $E_1, \dots, E_n$, $n \geq 2$, each with the Schur property, such that

$$Z = E_1 \hat{\otimes}_\pi \cdots \hat{\otimes}_\pi E_n$$

does not have the Schur property.

Rodriguez, *On weak compactness in projective tensor products*, arXiv:2206.08651, still records Schur preservation under projective tensor products as open. Thus this packet is not a full answer to Open Question 2; it proves that any negative answer to that tensor-product problem gives the desired proper inclusion.

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PROOF INTUITION

The composition ideal $\mathcal{CC} \circ \mathcal{L}$ can be detected on the linearization of a multilinear map: an n -linear operator belongs to $\mathcal{CC} \circ \mathcal{L}$ exactly when its associated linear operator on the projective tensor product is completely continuous. The canonical tensor map has identity as its linearization, so it fails to be in $\mathcal{CC} \circ \mathcal{L}$ precisely when the tensor product lacks the Schur property.

The remaining issue is whether the canonical tensor map lies in $\mathcal{L}_{msc}$. For Schur factors, weak convergence of elementary tensors is already norm convergence. The zero-limit case is forced by a subsequence/functionals argument using the Schur property in each factor; the nonzero case is reduced to the zero-limit case by splitting the factors into rank-one limit directions and complementary kernels.

ELEMENTARY TENSORS OVER SCHUR FACTORS

Lemma 1. *Let $E_1, \dots, E_n$ have the Schur property. If*

$$z_j = x_j^1 \otimes \cdots \otimes x_j^n$$

converges weakly to 0 in $E_1 \widehat{\otimes}_\pi \cdots \widehat{\otimes}_\pi E_n$, then

$$\|x_j^1\| \cdots \|x_j^n\| \longrightarrow 0.$$

Proof. The sequence (z_j) is weakly convergent, hence bounded. Suppose, after passing to a subsequence, that

$$\|x_j^1\| \cdots \|x_j^n\| \geq \delta > 0$$

for all j . Terms with a zero factor can be discarded, so put

$$u_j^k = x_j^k / \|x_j^k\| \quad (1 \leq k \leq n).$$

For every continuous n -linear form B ,

$$B(u_j^1, \dots, u_j^n) = \frac{B(x_j^1, \dots, x_j^n)}{\|x_j^1\| \cdots \|x_j^n\|} \longrightarrow 0.$$

Since E_1 has the Schur property and $\|u_j^1\| = 1$, the sequence (u_j^1) is not weakly null. Hence some subsequence and some $\varphi_1 \in E_1^*$ satisfy $|\varphi_1(u_j^1)| \geq \epsilon_1 > 0$ on that subsequence. Repeating this argument successively in $E_2, \dots, E_n$, we obtain a common subsequence and functionals $\varphi_k \in E_k^*$ such that $|\varphi_k(u_j^k)| \geq \epsilon_k > 0$ for every k along the common subsequence. The multilinear form

$$B(y_1, \dots, y_n) = \varphi_1(y_1) \cdots \varphi_n(y_n)$$

then does not tend to 0 on that subsequence, a contradiction. $\square$

Lemma 2. *Let $E_1, \dots, E_n$ have the Schur property. If elementary tensors*

$$z_j = x_j^1 \otimes \cdots \otimes x_j^n$$

converge weakly to an elementary tensor

$$z = x^1 \otimes \cdots \otimes x^n$$

in $E_1 \widehat{\otimes}_\pi \cdots \widehat{\otimes}_\pi E_n$, then $z_j \rightarrow z$ in projective norm.

Proof. If $z = 0$, this is Lemma 1. Suppose $z \neq 0$, so every $x^k \neq 0$. Choose $\varphi_k \in E_k^*$ with $\varphi_k(x^k) = 1$. Let $P_k(y) = \varphi_k(y)x^k$ and $Q_k = I_{E_k} - P_k$. For each subset $S \subseteq \{1, \dots, n\}$, let

$$R_S = T_1^S \widehat{\otimes}_\pi \cdots \widehat{\otimes}_\pi T_n^S, \quad T_k^S = \begin{cases} Q_k, & k \in S, \\ P_k, & k \notin S. \end{cases}$$

The operators R_S are bounded and weak-to-weak continuous on the projective tensor product. Also $R_\emptyset z = z$ and $R_S z = 0$ for $S \neq \emptyset$.

Since $z_j \rightarrow z$ weakly, $R_S z_j \rightarrow 0$ weakly for every nonempty S . Each $R_S z_j$ is an elementary tensor in a projective tensor product of closed subspaces of the E_k 's, hence again in Schur spaces. By Lemma 1, $\|R_S z_j\| \rightarrow 0$ for every nonempty S .

For $S = \emptyset$,

$$R_\emptyset z_j = \left(\prod_{k=1}^n \varphi_k(x_j^k) \right) x^1 \otimes \cdots \otimes x^n \longrightarrow z$$

in norm, because applying the scalar n -linear form $(y_1, \dots, y_n) \mapsto \prod_k \varphi_k(y_k)$ gives convergence of the coefficient to 1. The finite decomposition

$$z_j = \sum_{S \subseteq \{1, \dots, n\}} R_S z_j$$

therefore gives $z_j \rightarrow z$ in projective norm. $\square$

THE CONDITIONAL SEPARATION

Proposition 1. For $A \in \mathcal{L}(E_1, \dots, E_n; F)$, let $\tilde{A} : E_1 \widehat{\otimes}_\pi \cdots \widehat{\otimes}_\pi E_n \rightarrow F$ be its linearization. Then

$$A \in \mathcal{CC} \circ \mathcal{L}(E_1, \dots, E_n; F)$$

if and only if $\tilde{A}$ is completely continuous.

Proof. If $A = u \circ B$ with $u : G \rightarrow F$ completely continuous and $B : E_1 \times \cdots \times E_n \rightarrow G$ multilinear, then $\tilde{A} = u \circ \tilde{B}$. Since $\tilde{B}$ is weak-to-weak continuous and u sends weakly convergent sequences to norm-convergent sequences, $\tilde{A}$ is completely continuous.

Conversely, if $\tilde{A}$ is completely continuous, then

$$A = \tilde{A} \circ \sigma, \quad \sigma(x_1, \dots, x_n) = x_1 \otimes \cdots \otimes x_n,$$

is a $\mathcal{CC} \circ \mathcal{L}$ factorization. $\square$

Theorem 1. Assume the Schur tensor counterexample hypothesis. Then

$$\mathcal{CC} \circ \mathcal{L} \subsetneq \mathcal{L}_{msc}.$$

Proof. Let $E_1, \dots, E_n$ be Schur spaces such that

$$Z = E_1 \hat{\otimes}_\pi \cdots \hat{\otimes}_\pi E_n$$

does not have the Schur property. Consider the canonical tensor map

$$\sigma : E_1 \times \cdots \times E_n \rightarrow Z, \quad \sigma(x_1, \dots, x_n) = x_1 \otimes \cdots \otimes x_n.$$

If a sequence of tuples converges multilinearly, then the corresponding elementary tensors converge weakly in Z . By Lemma 2, they converge in projective norm. Hence $\sigma \in \mathcal{L}_{msc}(E_1, \dots, E_n; Z)$.

The linearization of σ is the identity operator I_Z . Since Z lacks the Schur property, I_Z is not completely continuous. By Proposition 1, $\sigma \notin \mathcal{CC} \circ \mathcal{L}(E_1, \dots, E_n; Z)$. Thus the inclusion is proper. $\square$

VERIFICATION NOTES

The proof above is complete modulo the explicitly stated Schur tensor counterexample hypothesis. No computational evidence is used. The most important review point is Lemma 2, especially the reduction from a nonzero elementary limit to the zero-limit lemma by tensoring the rank-one projections P_k and Q_k .

SEARCH BOUNDS

The run indexes were searched for arXiv:1504.00520, hyper-ideals, multilinearly sequentially continuous operators, $\mathcal{CC} \circ \mathcal{L}$, and projective tensor Schur phrases. No prior packet for this conditional reduction was found. Local parsed arXiv sources and a bounded web search on 2026-06-16 did not identify a later exact solution of the Schur tensor-product preservation problem; Rodriguez arXiv:2206.08651 still records it as open in 2022.

REFERENCES

- [1] G. Botelho and E. R. Torres, *Hyper-ideals of multilinear operators*, arXiv:1504.00520.
- [2] J. Rodriguez, *On weak compactness in projective tensor products*, arXiv:2206.08651.